CSP-203: Software Tools & Technologies Lab

Chapter – 1

Basic Linux Commands (Lecture – 1)

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What is Linux?

- Linux is an open-source Unix-like operating system.
- Popular for servers, development, and embedded systems.
- Interacted with through a **command-line interface (CLI)** or graphical interface.





From Wikipedia, the free encyclopedia

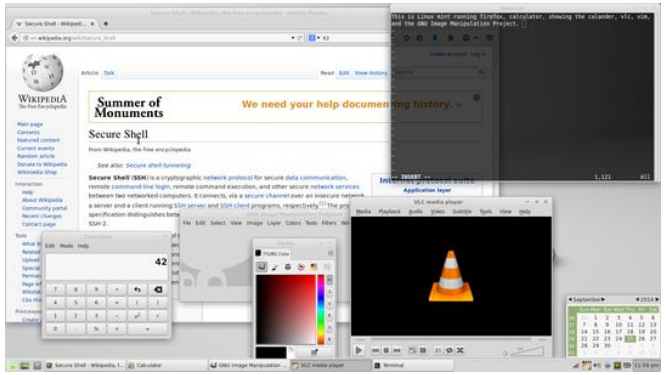
Unix (/ˈjuːnɪks/; trademarked as **UNIX**) is a family of multitasking, multi-user computer operating systems that derive from the original AT&T Unix, whose development started in 1969[1] at the Bell Labs research center by Ken Thompson, Dennis Ritchie, and others.[4]

Other Non-portable OS:
Written in assembly language and tightly coupled with specific hardware.

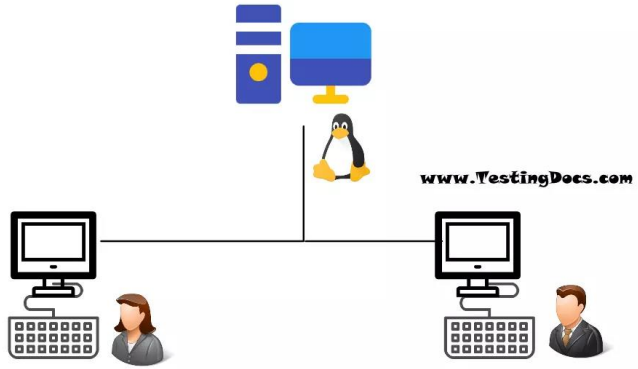
Unix distinguishes itself from its predecessors as the first portable operating system: almost the entire operating system is written in the C programming language, which allows Unix to operate on numerous platforms.[6]

Unix Features

1 Multitasking



2 Multiuser



3 Portable



4 Security

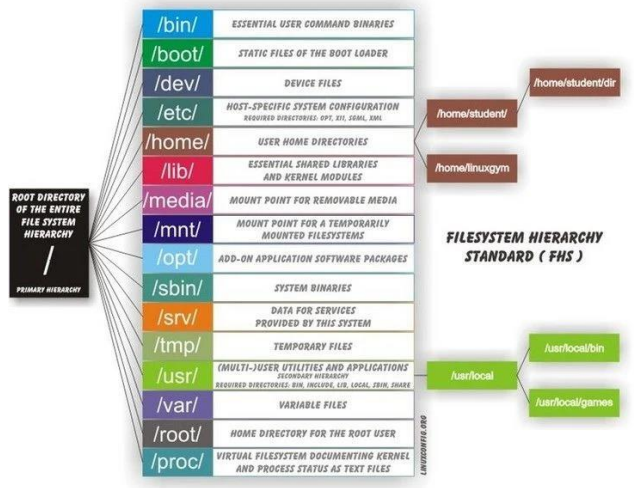


5 Utilities and commands *****

- manintro.ps, manintro.pdf, Title page and Introduction
- man11.ps, man11.pdf, Commands, part 1
- man12.ps, man12.pdf, Commands, part 2
- man13.ps, man13.pdf, Commands, part 3
- man14.ps, man14.pdf, Commands, part 4
- man21.ps, man21.pdf, System calls, part 1
- man22.ps, man22.pdf, System calls, part 2
- man31.ps, man31.pdf, Library routines
- man41.ps, man41.pdf, Special files
- man51.ps, man51.pdf, File formats
- man61.ps, man61.pdf, User-maintained software
- man71.ps, man71.pdf, Miscellaneous

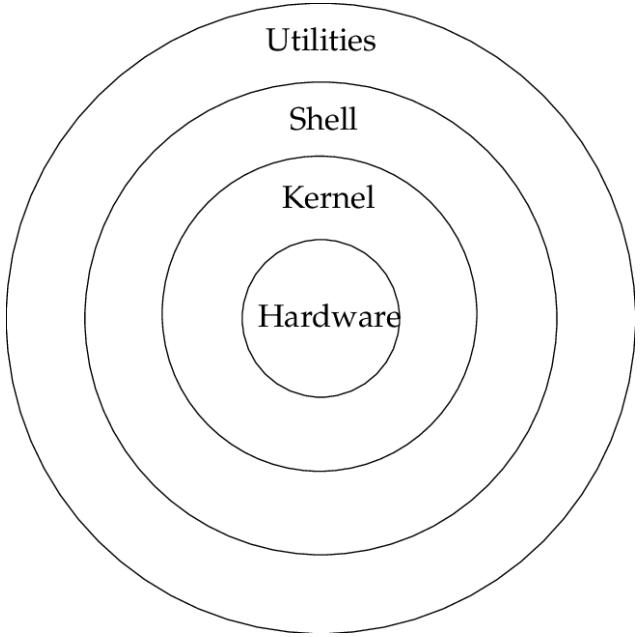
Unix Manual, first edition (bell-labs.com)

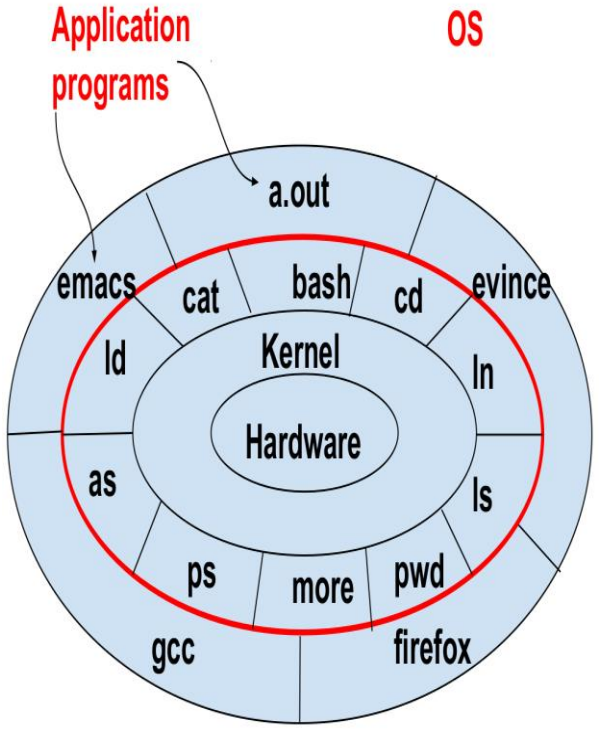
6 File and Directory *****



Hierarchy

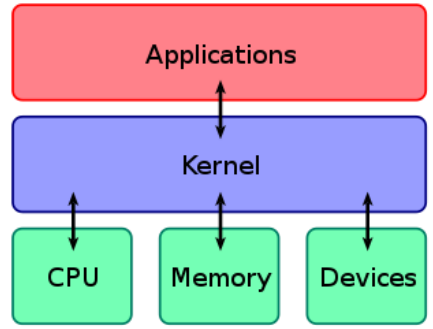
1. Kernel
2. Shell
3. Utilities and File Systems





https://minnie.tuhs.org/cgi-bin/utree.plierarchy

https://www.bell-labs.com/usr/dmr/www/hist.html



Shells

- A shell is a program that allows the user to interact with the Linux system:
  - Read the user's input and parse it
  - Evaluates special characters
  - Set up pipes, redirections, and background processing

  - Interprets and launches commands keyed-in at the command prompt
    `<username>@<hostname>:<pwd>{#|$}`

Types of Shells



We focus on bash: easy syntax and default in many systems

Stephen Bourne

Bill Joy

Davide Korn

Brain Fox

AT&T Bell Laboratories



AT&T Bell Laboratories

FREE SOFTWARE FOUNDATION

8 Types of Linux Shells (phoenixnap.com)

The birth of the Bash shell | Opensource.com

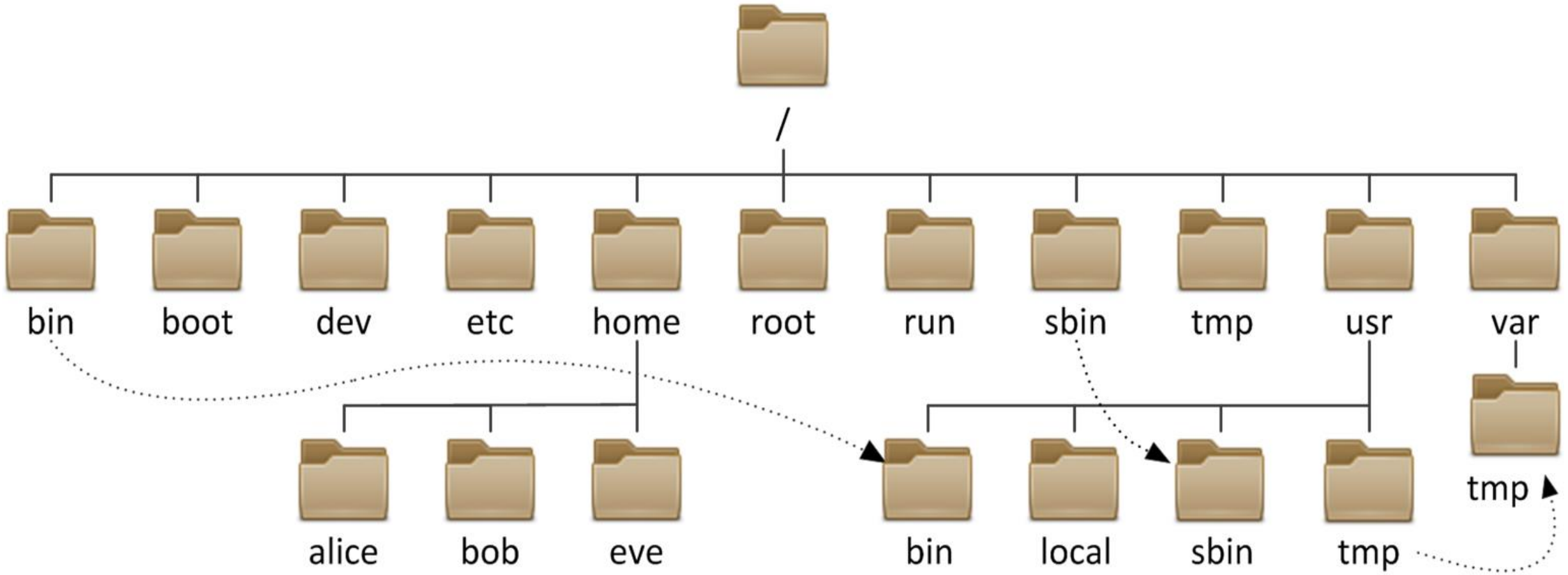
Comparison of command shells - Wikipedia

Linux Terminal



- root vs non-root (# vs $ prompt)
- Sudo su if you want to access as root user

File System Hierarchy



File system Hierarchy

- /dev: Hardware devices can be accessed here
- /lib: Stores libraries, along with /usr/lib, /usr/local/lib, etc.
- /mnt: Frequently used to mount disk drives
- /usr: Mostly user-installed programs and their related files
- /etc: System-wide settings
- /bin: System programs
- /usr/bin: Most user programs
- /usr/local/bin: A few other user programs

➢ What's a command?

- Programs that are keyed in by the user and launched by the shell

➢ Why Use the Command Line?

- Powerful and flexible
- Automate tasks via scripts
- Use system tools efficiently
- Access features not available in GUI

Types of Commands

Types of commands

- **Internal (built-in) commands:** Internal (built-in) commands are programs implemented by the shell itself
  - Example: echo, cd, pwd, exit, help, type

- **External commands:**
  - External commands are programs (mostly c) in /bin or /usr/bin or /usr/sbin directory
  - TO know the location of a command, use ***which*** or ***whereis*** command

```
$ type cd
cd is a shell builtin

$ type ls
mkdir is /bin/ls


$ which cd
$ which type

$ whereis cd
cd:


 $ which ls
 /bin/ls

 $ which git
 /usr/bin/git

 $ whereis ls
 ls: /bin/ls
 /usr/share/man/man1/ls.1.gz
```

Manual for each coomand

- man <command_name>
  - Bring up the manual page (manpage) for the selected command
  - Unlike Web search, manpages are system-specific
  - Gives a comprehensive list of all possible options/parameters
  - Use /<keyword> to perform a keyword search in a manpage
  - The n-key jumps to successive search results

Types of Commands

File and file system management:

cd | pwd | mkdir | touch | ls | cp | mv | rm | rmdir | which | info | df | du | chmod | chown | chgrp | locate | find | ln |quota

General User Commands:

env | id | passwd | su | sudo | uname | who | whoami | cal | date | clear | history

Process management:

ps | top | kill | killall | pidof | pkill | ps | sleep | time | wait | watch

Text processing:

echo | cat | diff | cmp | head | tail | join | less | more | cut| paste | sed | sort | grep | awk | wc

N/w Communications:

ping | ifconfig | ip -a | netstat | traceroute | ssh | wget

Filesystem Utilities

cd – Change to another directory location

pwd – Print the current working directory

mkdir newdir – Make a directory

touch file.txt– Create a new file or update its modification time

ls / ls -l – List directory contents/ View permissions and ownership

cp file1.txt file2.txt – Copy a file or directory to another location

mv file.txt /path/ – Move or rename a file or directory

rm file.txt – Delete a file or directory tree

rmdir – Delete an empty directory

rm -r dir/ – Remove a directory and its contents

which - locate a command location

info – The GNU alternative to man

man – View manual pages for commands

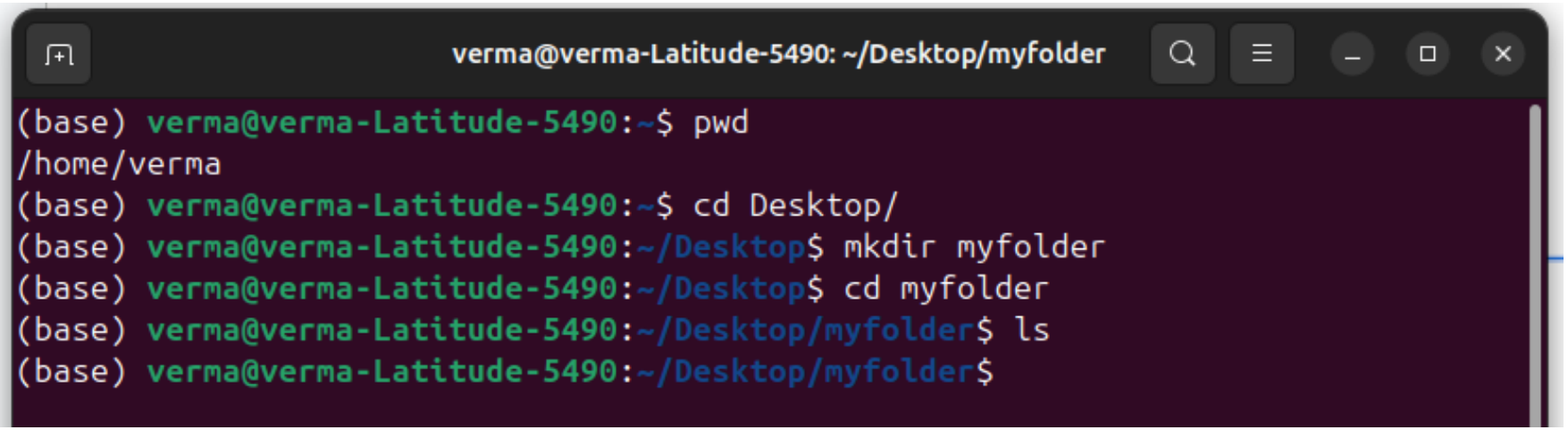
1. Directory Navigation

pwd # Show current directory

cd desktop # Change directory

mkdir myfolder # Create a new folder

cd myfolder # Enter the folder



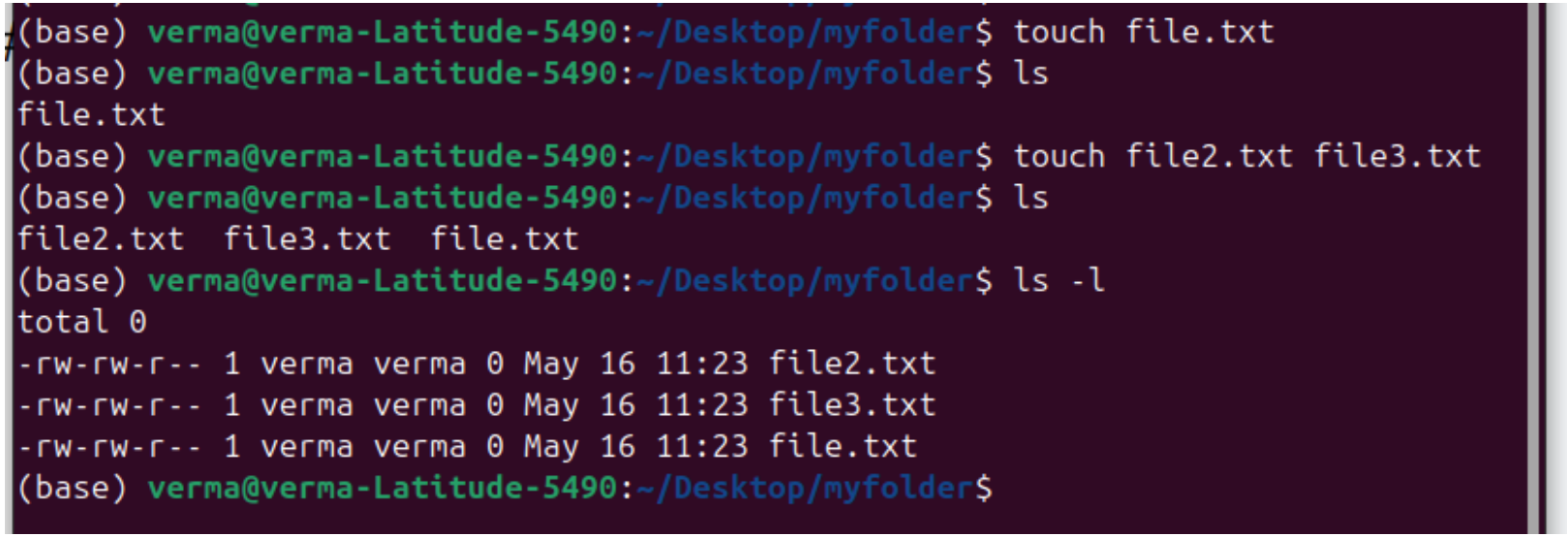
2. File Creation & Listing

touch file1.txt # Create files

touch file2.txt file3.txt # Create multiple files once

ls # List files

ls -l # Detailed listing



3. Copy, Move, Rename, Delete

```
cp file1.txt copy.txt                  # Copy file

mv copy.txt file4.txt                  # Rename file

rm file4.txt                           # Remove file

rmdir emptyfoldername                  # Remove empty directory

rm -r foldername                       # Remove a Folder with Contents
(Files/Subfolders)
```

Demo:

```
(base) verma@verma-Latitude-5490:~/Desktop/myfolder$ ls
file2.txt  file3.txt  file.txt
(base) verma@verma-Latitude-5490:~/Desktop/myfolder$ cp file2.txt copy.txt
(base) verma@verma-Latitude-5490:~/Desktop/myfolder$ ls
copy.txt  file2.txt  file3.txt  file.txt
(base) verma@verma-Latitude-5490:~/Desktop/myfolder$ mv copy.txt file4.txt
(base) verma@verma-Latitude-5490:~/Desktop/myfolder$ ls
file2.txt  file3.txt  file4.txt  file.txt
(base) verma@verma-Latitude-5490:~/Desktop/myfolder$ rm file4.txt
(base) verma@verma-Latitude-5490:~/Desktop/myfolder$ ls
file2.txt  file3.txt  file.txt
(base) verma@verma-Latitude-5490:~/Desktop/myfolder$ mkdir newfolder
(base) verma@verma-Latitude-5490:~/Desktop/myfolder$ ls
file2.txt  file3.txt  file.txt  newfolder
(base) verma@verma-Latitude-5490:~/Desktop/myfolder$ rmdir newfolder/
(base) verma@verma-Latitude-5490:~/Desktop/myfolder$ ls
file2.txt  file3.txt  file.txt
(base) verma@verma-Latitude-5490:~/Desktop/myfolder$ mkdir testdir
(base) verma@verma-Latitude-5490:~/Desktop/myfolder$ touch testdir/file1.txt
(base) verma@verma-Latitude-5490:~/Desktop/myfolder$ ls
file2.txt  file3.txt  file.txt  testdir
(base) verma@verma-Latitude-5490:~/Desktop/myfolder$ rm -r testdir
(base) verma@verma-Latitude-5490:~/Desktop/myfolder$ ls
file2.txt  file3.txt  file.txt
(base) verma@verma-Latitude-5490:~/Desktop/myfolder$
```

4. Command Location and Help

```
which cp          # Show path of command
man mv            # Manual page
info rm          # GNU help page
```

Filesystem Utilities

chmod – Change the permissions of a file or directory

Usertype – user(u) , group(g) , others(o)

Permission type – Read(r) , write (w) , execute(x)

Eg. chmod 755 file.sh # rwx for user, rx for group and others

chgrp – Change the group of a file or directory

chown – Change the owner of a file or directory

locate - find files by name

ln – Link one file/directory to another

strings - print the strings of printable characters in files.

find – Search for files through a directory hierarchy

quota – display disk usage and limits

Permissions and Ownership

We can use ls -l to tell us about ownership and permissions of files
ls -l - lists files and directories in the long format

Example:
-rw-rw-r-- 1 verma verma 0 Aug 16 11:23 file2.txt

rwxrwxrwx
User's Permissions
Group's Permissions
Other's permissions
R = Read, W = Write, X = Execute